Game Theory

Studies decision making in strategic environments
- began by studying actual games
- now used in nearly all areas of economics
  - firms
  - consumers
  - governments

a "Game Theoretic" model

1) Players: decision makers
2) Actions: available choices for each player
3) Timing: order of choices
4) Information Structure: what players know when they are making their choice
5) Payoffs: what each player gets (utility) at the end

note: payoffs depend on actions of all players

Equilibrium: Nash Equilibrium ← John Nash (1950), also Cournot!

Each player chooses a strategy that gives them the highest payoff given the strategies of other players

→ each player plays a "best response"

The Prisoners' Dilemma

assumption: prisoners don't care about eachother and want less jail time

Outcomes: 10 years ⟶ payoff 0
5 years ⟶ payoff 1
1 year ⟶ payoff 2
free ⟶ payoff 3

Payoff Matrix

prisoner 1 \ prisoner 2	confess	quiet
confess	1, 1	3, 0
quiet	0, 3	2, 2

notes: if the other player confesses → best response is confess
if the other player stays quiet → best response is confess

equilibrium: both players confess
→ both would be better off if they could both stay quiet but neither has incentive to do this

→ self interest is not aligned with social interest

ex. pollution, arms races, political attack ads, ...

Ex. Cartels (again)

Cournot

firm 1 \ firm 2	compete	collude
compete	55, 55	120, 40
collude	40, 120	100, 100

Cartel

Here, "compete" is a dominant strategy
→ gives a relatively higher payoff in each case of what the other player does

"collude" is a dominated strategy

Sometimes we can find the equilibrium of a game just by eliminating dominated strategies

Iterated Elimination of Dominated Strategies

prof \ student	no study	med.	hard	
easy	0,10	2,9	3,8	X_1
med	1,1	5,8	$\boxed{8,9}$	
hard	2,4	4,7	7,5	X_3
	X_2	X_4		

1. easy is dominated by medium for prof → eliminate easy
2. now no studying is dominated by medium (or hard) for student → eliminate no studying
3. now hard exam is dominated by prof → eliminate hard
4. now medium is dominated for student → eliminate medium

Equilibrium: $\left(\begin{matrix}\text{medium} \\ \text{exam}\end{matrix}, \begin{matrix}\text{hard} \\ \text{studying}\end{matrix}\right)$

Sometimes you can't solve a game this way though

ex. Bank Runs: a fear of insolvency causes a large number of depositors to withdraw their money simultaneously

depositor 1 \ depositor 2	withdraw	don't
withdraw	25,25*	50,0
don't	0,50	110,110*

neither player has a dominated strategy

best respones: if ② doesn't withdraw → don't
if ② withdraws → withdraw

there are 2 Nash Equilibria in this model